

L2.3 How Derivatives Affect the Shape of a Graph

Work in your group. Check your answers against the Quarto tonight.

The I/D Test and the sign test

1. On which intervals is f increasing, and on which is it decreasing?

$$f(x) = x^2 - 6x + 5$$

Stop. Write the sign of f' on each piece, and write down the number you evaluated to get it.

2. Same question, one more critical number.

$$f(x) = 2x^3 - 3x^2 - 12x$$

The First Derivative Test

3. Find the critical numbers of f , then find every local extremum.

$$f(x) = x^{2/3}$$

Stop. f' is never zero here. There is still a critical number — what is the other way to get one?

4. $f(x) = (x - 2)^3$ has exactly one critical number. Find it, and say what — if anything — is there.

Concavity and the Second Derivative Test

5. For $f(x) = x^4 - 4x + 1$: find every PIP, then find the intervals of concavity. For each PIP, say whether it is an inflection point, and **why**.

6. Use the **Second** Derivative Test to classify the critical numbers of

$$f(x) = 2x^3 + 3x^2 - 36x$$

7. (a) Find the local extrema of $f(x) = x^4 - 8x^2$ using **both** the First and the Second Derivative Test. Which one did you prefer, and why?
- (b) Now try the Second Derivative Test on $g(x) = (x - 1)^4$ at its critical number. What does it tell you? Settle the question another way.

Stop. One of these tests can come back with no verdict at all. That is not the same as a verdict of *no extremum*.

The derivatives table

8. Consider the function

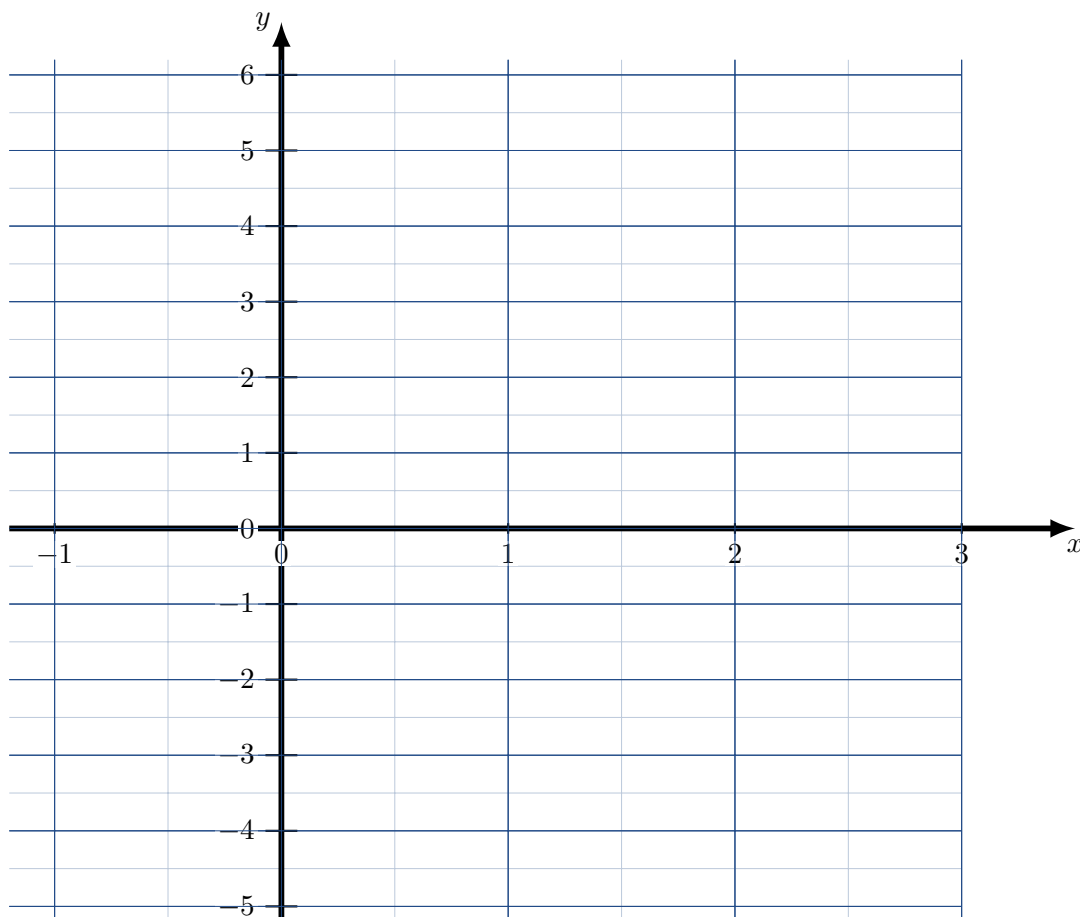
$$f(x) = 2x^3 - 9x^2 + 12x - 3$$

- (a) Find the intervals on which f is increasing or decreasing.
- (b) Find any local minimum and Maximum values of f .
- (c) Find the intervals of concavity and the inflection points.

9. Same function. Find the y -intercept, and use that the only x -intercept is ≈ 0.322 . Then **diagram the function** using the method from the top of the period.

Stop. Every CN and every PIP goes on the *same* number line, and the pieces are what is left between all of them.

10. Same function. Sketch it on the axes below, reading direction off the arrows and bend off the cups.



To check it, load this function into the table builder in the Quarto and step through it.

11. Find the CNs, extrema and IPs of

$$f(x) = \frac{2}{x^2 + 1}$$

See if you can sketch it.

Stop. The quotient rule twice is the long road. Rewrite it as $2(x^2 + 1)^{-1}$ first.

12. **BOARDS** Group work. A function is defined by

$$f(x) = x^3 - 2x^2 - 3x - 1$$

- (a) Graph $f(x)$. Label any inflection points (IP), any local Max points (M) and any local min points (m).
- (b) Graph $f'(x)$ on the same axes. What happened to the IPs?
- (c) Graph $f''(x)$ on the same axes. What happened to the Ms and ms?
- (d) Complete the sentence. IPs become _____ in the graph of the first derivative of a function, which become _____ in the graph of the second derivative.

Then do the same for $g(x) = x^3 - 3x - 1$ and explain what changed.